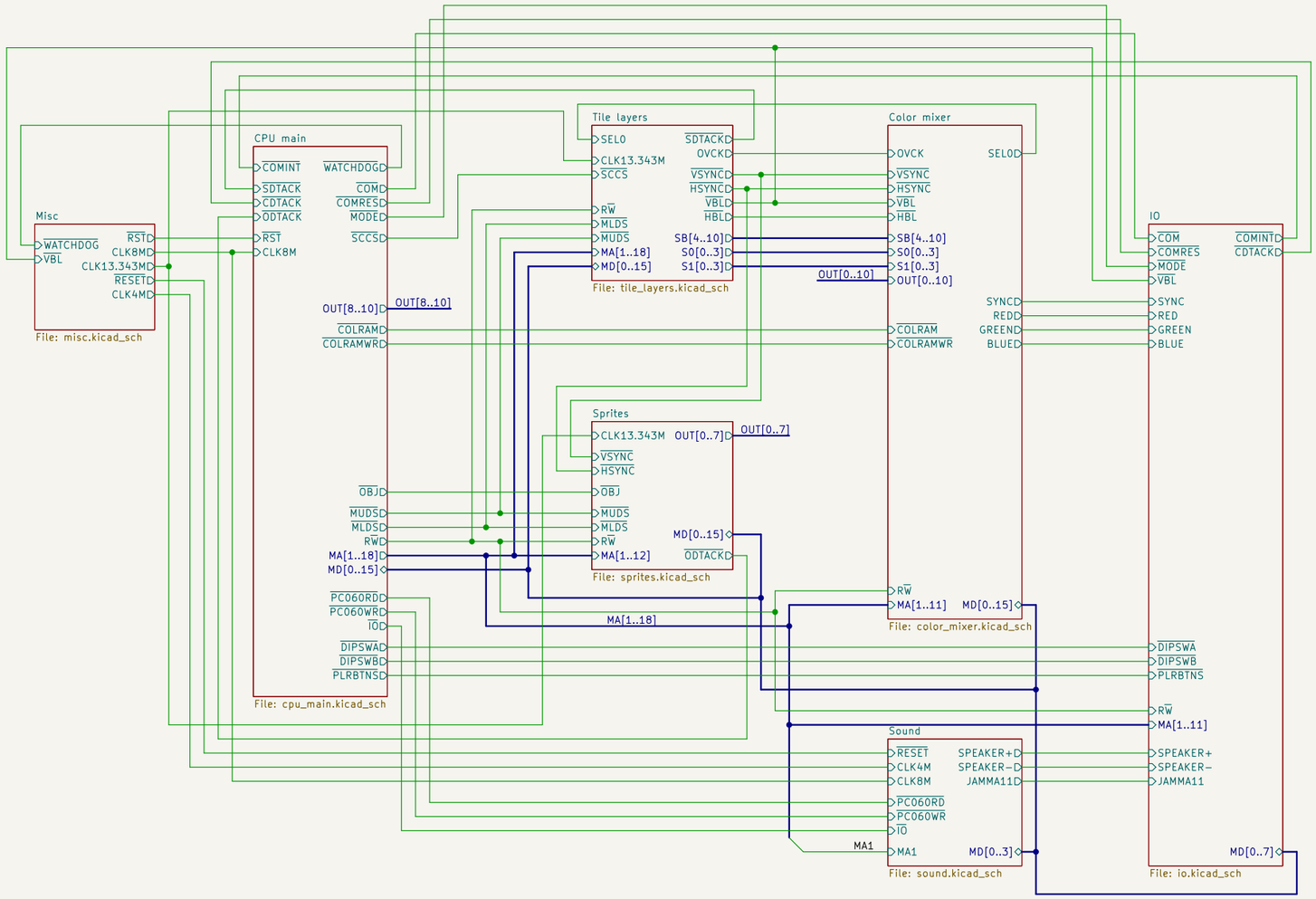


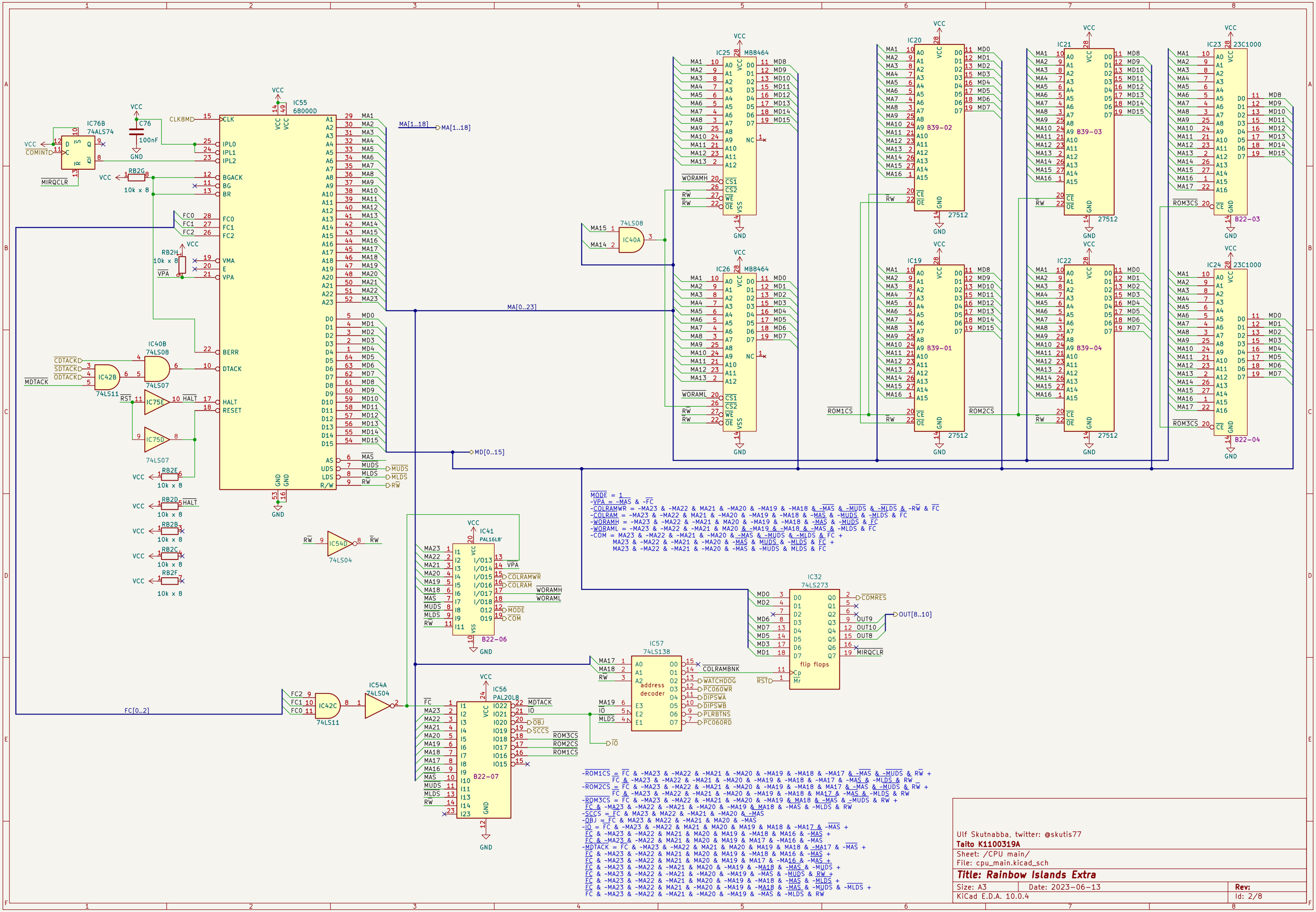
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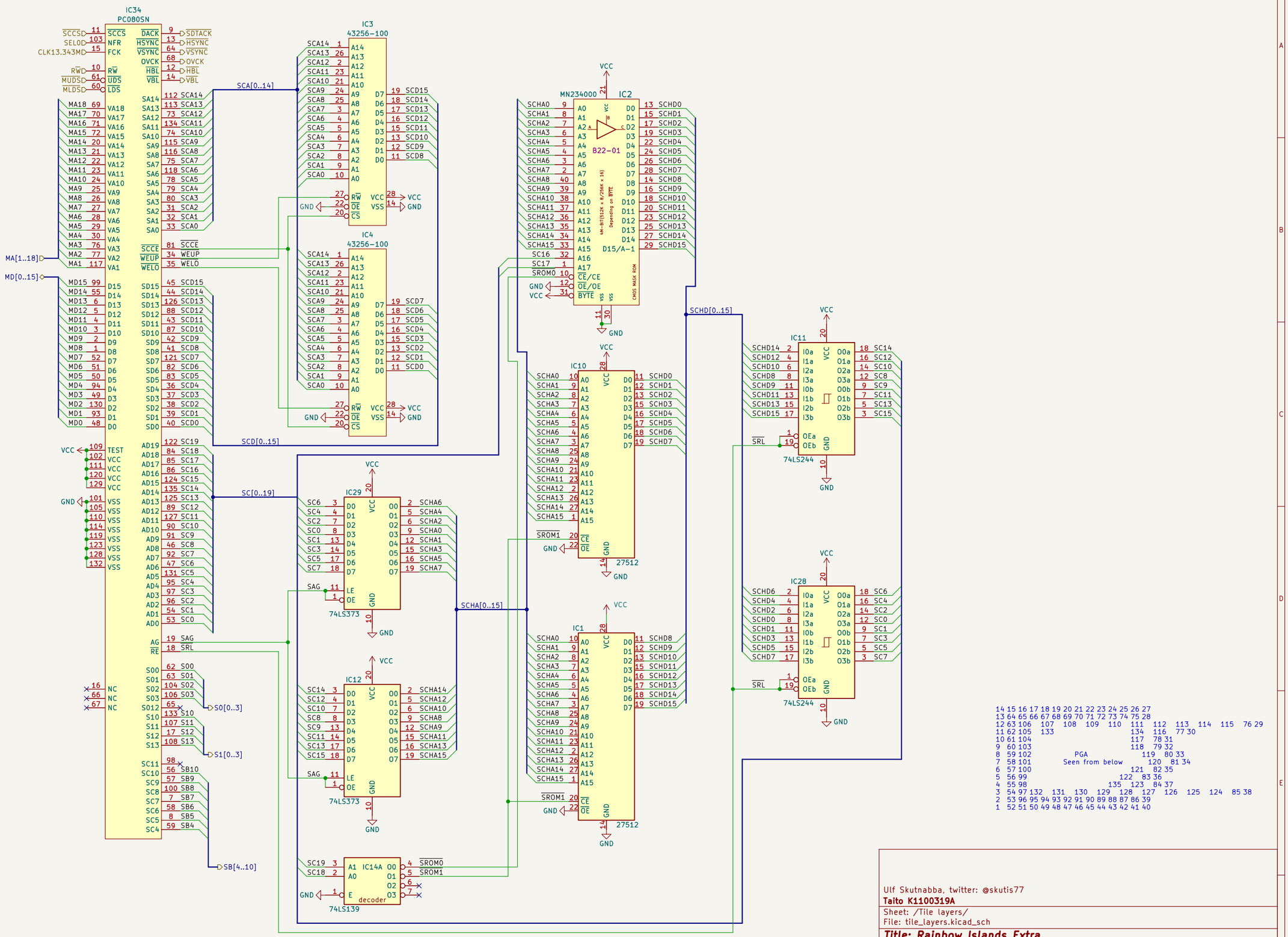
Taito K1100319A

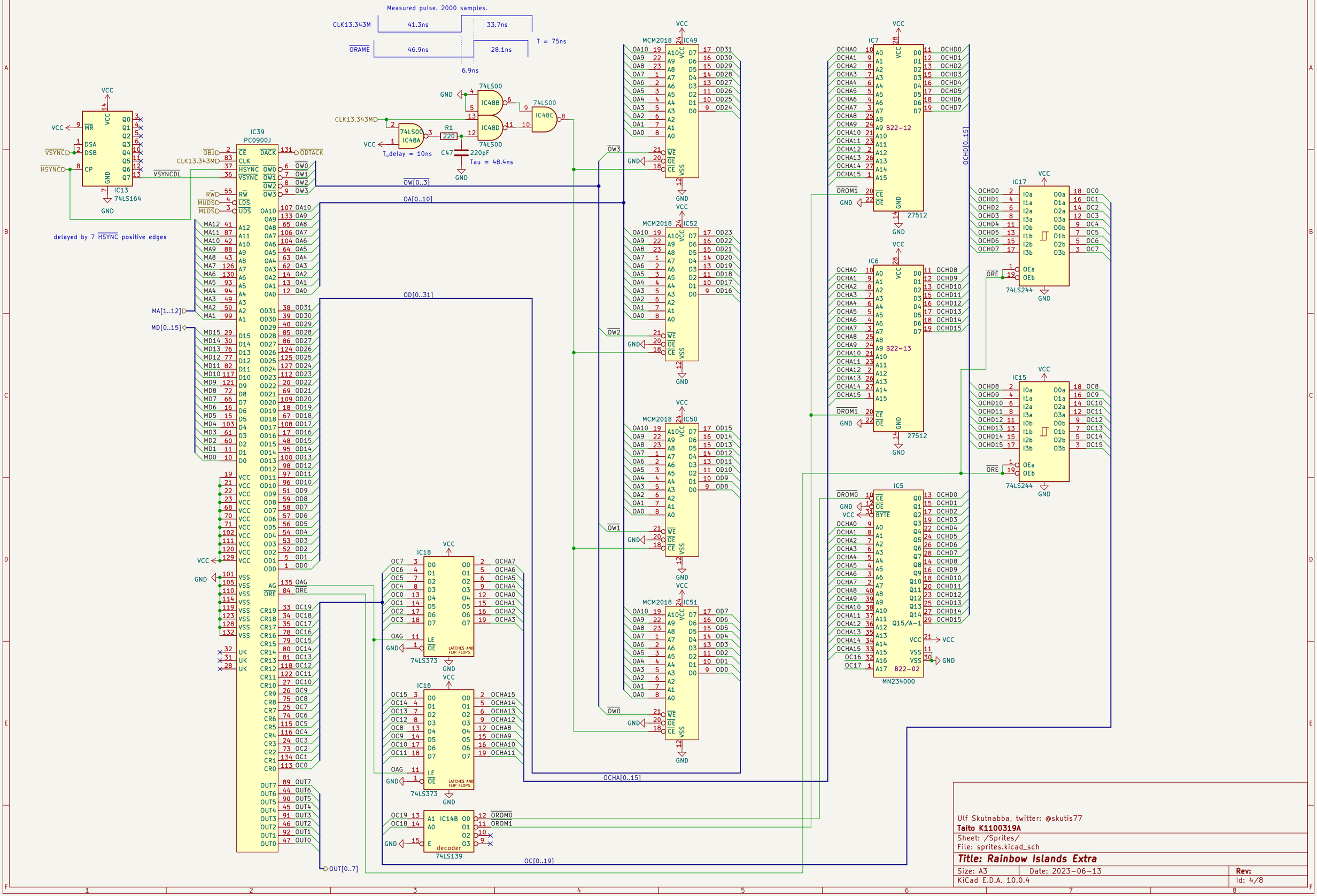
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File: risle.kicad_sch

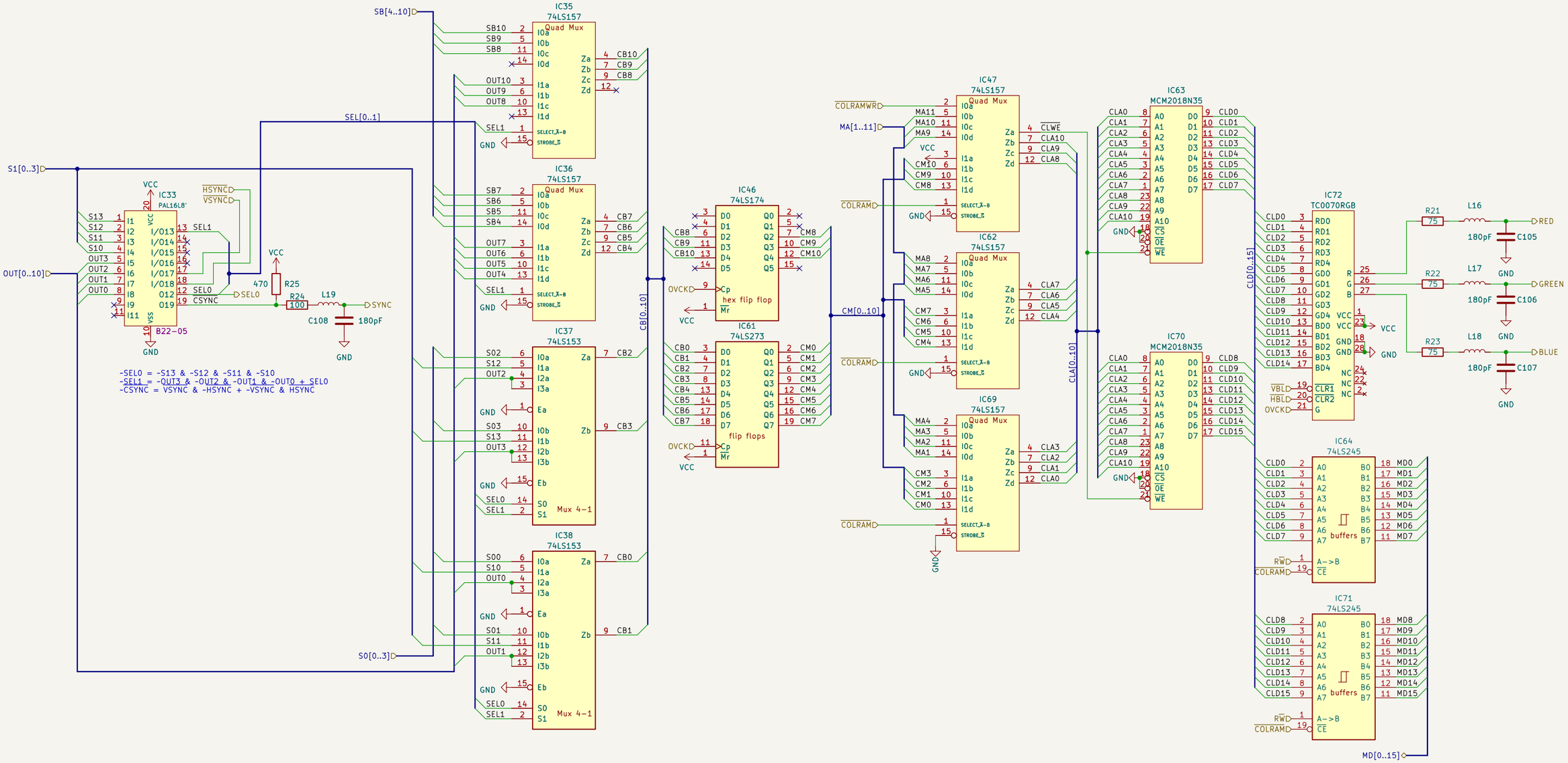
Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
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IC55
68000D
IC76B
74ALS74
C76
100nF
RB2G
10k x 8
RB2H
RB2E
RB2D
RB2B
RB2C
RB2F
IC40B
74LS08
IC42B
74LS11
74LS07
IC75E
IC75D
IC54B
74LS04
IC54D
IC54A
IC42C
IC25 MB8464
IC26 MB8464
IC20 B39-02
IC19 B39-01
IC21 B39-03
IC22 B39-04
IC23 23C1000
IC24 23C1000
27512
B22-03
B22-04
IC40A
IC41 PAL16L8'
B22-06
IC56 PAL20L8
B22-07
IC57 74LS138
address decoder
IC32 74LS273
flip flops
MA[1..18]
MA[0..23]
MD[0..15]
FC[0..2]
OUT[8..10]
MODE = 1
-VPA = -MAS & -FC
-COLRAMWR = -MA23 & -MA22 & MA21 & -MA20 & -MA19 & -MA18 & -MAS & -MUDS & -MLDS & -RW & FC
-COLRAM = -MA23 & -MA22 & MA21 & -MA20 & -MA19 & -MA18 & -MAS & -MUDS & -MLDS & FC
-WORAMH = -MA23 & -MA22 & -MA21 & MA20 & -MA19 & -MA18 & -MAS & -MUDS & FC
-WORAML = -MA23 & -MA22 & -MA21 & MA20 & -MA19 & -MA18 & -MAS & -MLDS & FC
-COM = MA23 & -MA22 & -MA21 & -MA20 & -MAS & -MUDS & -MLDS & FC +
MA23 & -MA22 & -MA21 & -MA20 & -MAS & MUDS & -MLDS & FC +
MA23 & -MA22 & -MA21 & -MA20 & -MAS & -MUDS & MLDS & FC
-ROM1CS = FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MA18 & -MA17 & -MAS & -MUDS & RW +
FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MA18 & -MA17 & -MAS & -MLDS & RW
-ROM2CS = FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MA18 & MA17 & -MAS & -MUDS & RW +
FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MA18 & MA17 & -MAS & -MLDS & RW
-ROM3CS = FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & MA18 & -MAS & -MUDS & RW +
FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & MA18 & -MAS & -MLDS & RW
-SCCS = FC & MA23 & MA22 & -MA21 & -MA20 & -MAS
-OBJ = FC & MA23 & MA22 & -MA21 & MA20 & -MAS
-IO = FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & MA18 & -MA17 & -MAS +
FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & -MA18 & MA16 & -MAS +
FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & MA17 & -MA16 & -MAS
-MDTACK = FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & MA18 & -MA17 & -MAS +
FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & -MA18 & MA16 & -MAS +
FC & -MA23 & -MA22 & MA21 & MA20 & MA19 & MA17 & -MA16 & -MAS +
FC & -MA23 & -MA22 & -MA21 & MA20 & -MA19 & -MA18 & -MAS & -MUDS +
FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MAS & -MUDS & RW +
FC & -MA23 & -MA22 & -MA21 & MA20 & -MA19 & -MA18 & -MAS & -MLDS +
FC & -MA23 & -MA22 & MA21 & -MA20 & -MA19 & -MA18 & -MAS & -MUDS & -MLDS +
FC & -MA23 & -MA22 & -MA21 & -MA20 & -MA19 & -MAS & -MLDS & RW
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Sheet: /CPU main/
File: cpu_main.kicad_sch
Title: Rainbow Islands Extra
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IC34
PC080SN
IC3
43256-100
IC4
43256-100
IC2
MN234000
B22-01
IC10
IC1
27512
IC11
74LS244
IC28
74LS244
IC29
74LS373
IC12
74LS373
IC14A
74LS139
decoder
SCA[0..14]
SCD[0..15]
SC[0..19]
SCHA[0..15]
SCHD[0..15]
MA[1..18]
MD[0..15]
S0[0..3]
S1[0..3]
SB[4..10]
VCC
GND
SRL
14 15 16 17 18 19 20 21 22 23 24 25 26 27
13 64 65 66 67 68 69 70 71 72 73 74 75 28
12 63 106 107 108 109 110 111 112 113 114 115 76 29
11 62 105 133 134 116 77 30
10 61 104 117 78 31
9 60 103 118 79 32
8 59 102 PGA 119 80 33
7 58 101 Seen from below 120 81 34
6 57 100 121 82 35
5 56 99 122 83 36
4 55 98 135 123 84 37
3 54 97 132 131 130 129 128 127 126 125 124 85 38
2 53 96 95 94 93 92 91 90 89 88 87 86 39
1 52 51 50 49 48 47 46 45 44 43 42 41 40
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Sheet: /Tile layers/
File: tile_layers.kicad_sch
Title: Rainbow Islands Extra
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Measured pulse. 2000 samples.
CLK13.343M
41.3ns
33.7ns
T = 75ns
ORAME
46.9ns
28.1ns
6.9ns
IC13
74LS164
delayed by 7 HSYNC positive edges
IC39
PC0900J
VSYNCDL
MA[1..12]
MD[0..15]
OBJ
CLK13.343M
HSYNC
VSYNC
RW
MUDS
MLDS
ODTACK
OW0
OW1
OW2
OW3
OA[0..10]
OD[0..31]
OAG
ORE
OUT[0..7]
74LS00
IC48A
IC48B
IC48C
IC48D
R1
220
C47
220pF
Tau = 48.4ns
T_delay = 10ns
MCM2018
IC49
IC52
IC50
IC51
OW[0..3]
IC7
IC6
IC5
B22-12
B22-13
B22-02
27512
MN234000
OCHD[0..15]
IC17
IC15
74LS244
IC18
IC16
74LS373
LATCHES AND FLIP FLOPS
IC14B
decoder
74LS139
OCHA[0..15]
OC[0..19]
OROM0
OROM1
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Taito K1100319A
Sheet: /Sprites/
File: sprites.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
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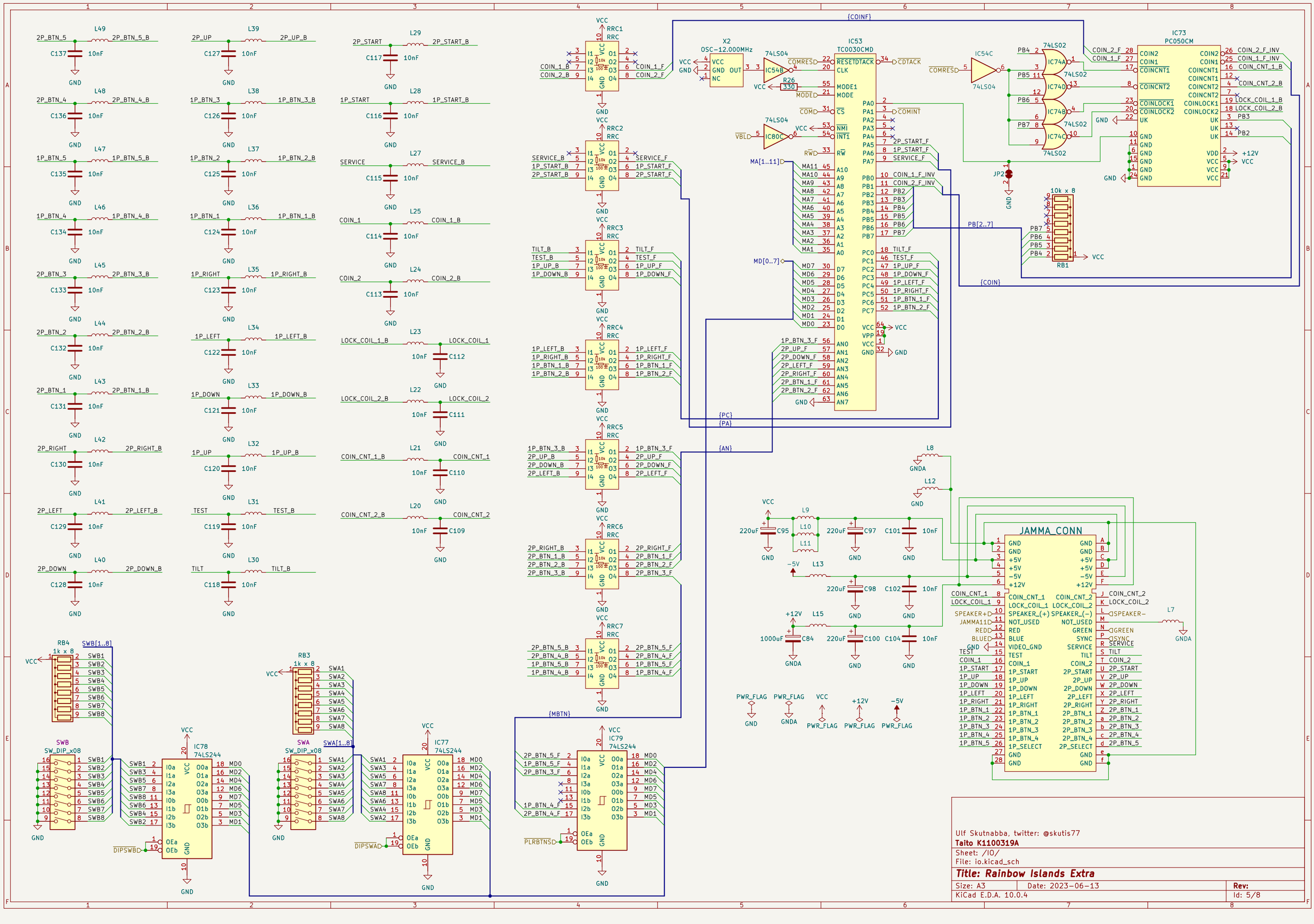
Taito K1100319A

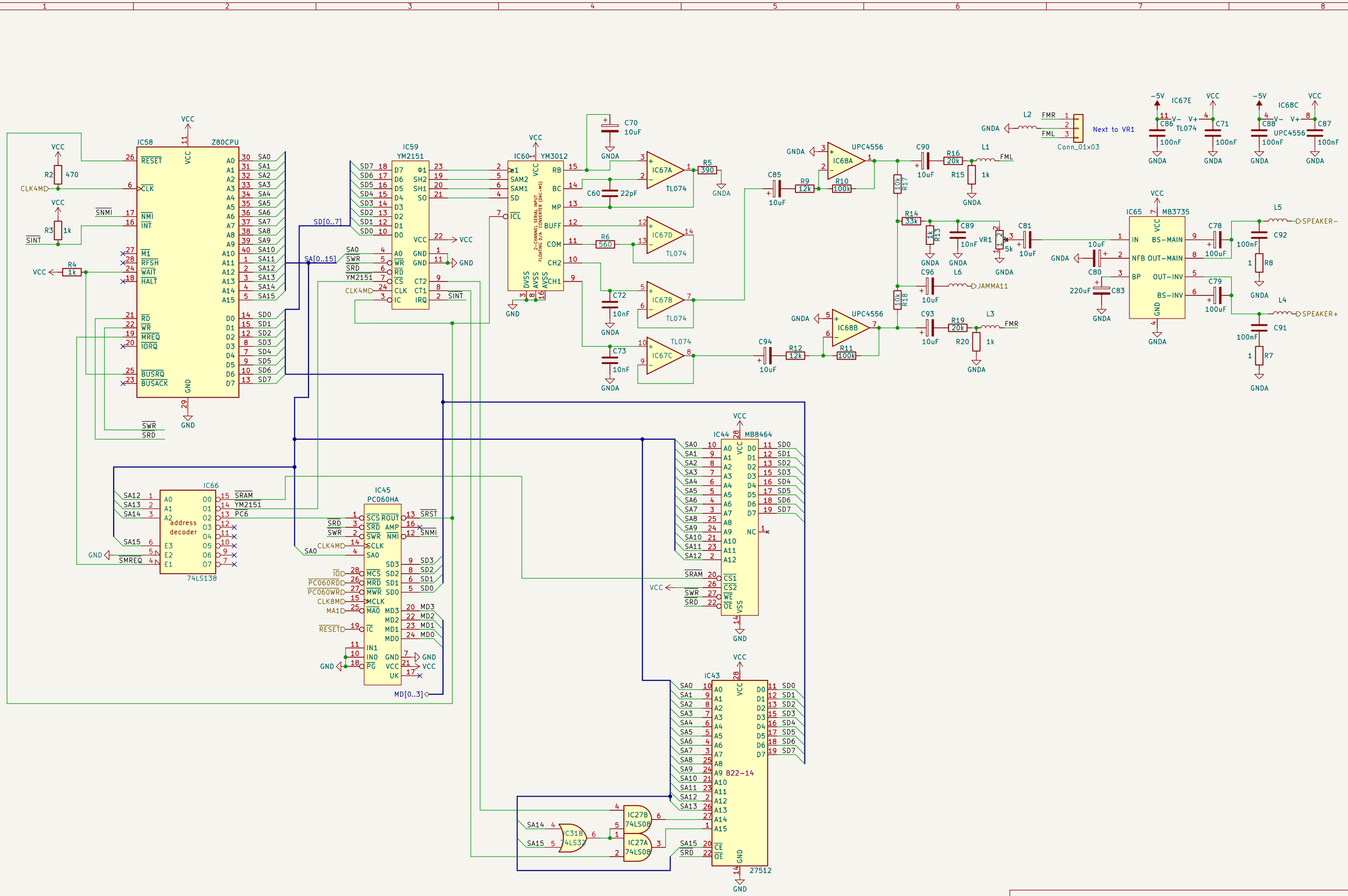
Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Rainbow Islands Extra

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2P_BTN_5
L49
2P_BTN_5_B
C137
10nF
GND
2P_BTN_4
L48
2P_BTN_4_B
C136
1P_BTN_5
L47
1P_BTN_5_B
C135
1P_BTN_4
L46
1P_BTN_4_B
C134
2P_BTN_3
L45
2P_BTN_3_B
C133
2P_BTN_2
L44
2P_BTN_2_B
C132
2P_BTN_1
L43
2P_BTN_1_B
C131
2P_RIGHT
L42
2P_RIGHT_B
C130
2P_LEFT
L41
2P_LEFT_B
C129
2P_DOWN
L40
2P_DOWN_B
C128
2P_UP
L39
2P_UP_B
C127
1P_BTN_3
L38
1P_BTN_3_B
C126
1P_BTN_2
L37
1P_BTN_2_B
C125
1P_BTN_1
L36
1P_BTN_1_B
C124
1P_RIGHT
L35
1P_RIGHT_B
C123
1P_LEFT
L34
1P_LEFT_B
C122
1P_DOWN
L33
1P_DOWN_B
C121
1P_UP
L32
1P_UP_B
C120
TEST
L31
TEST_B
C119
TILT
L30
TILT_B
C118
2P_START
L29
2P_START_B
C117
1P_START
L28
1P_START_B
C116
SERVICE
L27
SERVICE_B
C115
COIN_1
L25
COIN_1_B
C114
COIN_2
L24
COIN_2_B
C113
LOCK_COIL_1_B
L23
LOCK_COIL_1
C112
LOCK_COIL_2_B
L22
LOCK_COIL_2
C111
COIN_CNT_1_B
L21
COIN_CNT_1
C110
COIN_CNT_2_B
L20
COIN_CNT_2
C109
VCC
RRC1
RRC
COIN_1_B
COIN_2_B
COIN_1_F
COIN_2_F
RRC2
SERVICE_B
1P_START_B
2P_START_B
SERVICE_F
1P_START_F
2P_START_F
RRC3
TILT_B
TEST_B
1P_UP_B
1P_DOWN_B
TILT_F
TEST_F
1P_UP_F
1P_DOWN_F
RRC4
1P_LEFT_B
1P_RIGHT_B
1P_BTN_1_B
1P_BTN_2_B
1P_LEFT_F
1P_RIGHT_F
1P_BTN_1_F
1P_BTN_2_F
RRC5
1P_BTN_3_B
2P_UP_B
2P_DOWN_B
2P_LEFT_B
1P_BTN_3_F
2P_UP_F
2P_DOWN_F
2P_LEFT_F
RRC6
2P_RIGHT_B
2P_BTN_1_B
2P_BTN_2_B
2P_BTN_3_B
2P_RIGHT_F
2P_BTN_1_F
2P_BTN_2_F
2P_BTN_3_F
RRC7
2P_BTN_5_B
2P_BTN_4_B
1P_BTN_5_B
1P_BTN_4_B
2P_BTN_5_F
2P_BTN_4_F
1P_BTN_5_F
1P_BTN_4_F
{COINF}
{PC}
{PA}
{AN}
{MBTN}
{COIN}
X2
OSC-12.000MHz
IC53
TC0030CMD
RESETDTACK
CLK
MODE1
MODE
CS
NMI
INT1
RW
CDTACK
COMRES
COMINT
VBLL
MA[1..11]
MD[0..7]
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
AN0
AN1
AN2
AN3
AN4
AN5
AN6
AN7
74LS04
IC54B
IC80C
IC54C
74LS02
IC74A
IC74B
IC74C
IC74D
JP2
10k x 8
RB1
PB[2..7]
IC73
PC050CM
COIN2
COIN1
COINCNT1
COINCNT2
COINLOCK1
COINLOCK2
UK
COIN2_F
COIN1_F
COIN_CNT_1_B
COIN_CNT_2_B
LOCK_COIL_1_B
LOCK_COIL_2_B
PB3
PB2
+12V
RB4
1k x 8
SWB[1..8]
SWB
SW_DIP_x08
IC78
74LS244
DIPSWB
RB3
SWA
SWA[1..8]
IC77
DIPSWA
IC79
PLRBTNS
L8
GNDA
L12
L9
L10
L11
L13
L15
L7
220uF
C95
C97
C101
C98
C102
C84
C100
C104
10nF
1000uF
JAMMA_CONN
SPEAKER+
SPEAKER-
JAMMA11
RED
GREEN
BLUE
SYNC
SERVICE
VIDEO_GND
TEST
COIN_1
1P_START
1P_UP
1P_DOWN
1P_LEFT
1P_RIGHT
1P_BTN_1
1P_BTN_2
1P_BTN_3
1P_BTN_4
1P_SELECT
2P_START
2P_UP
2P_DOWN
2P_LEFT
2P_RIGHT
2P_BTN_1
2P_BTN_2
2P_BTN_3
2P_BTN_4
2P_SELECT
PWR_FLAG
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Sheet: /IO/
File: io.kicad_sch
Title: Rainbow Islands Extra
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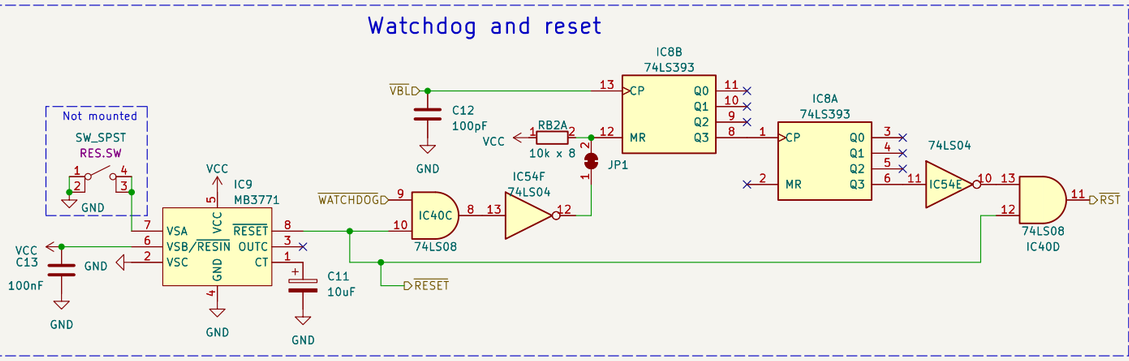
Sheet: /Sound/

File: sound.kicad_sch

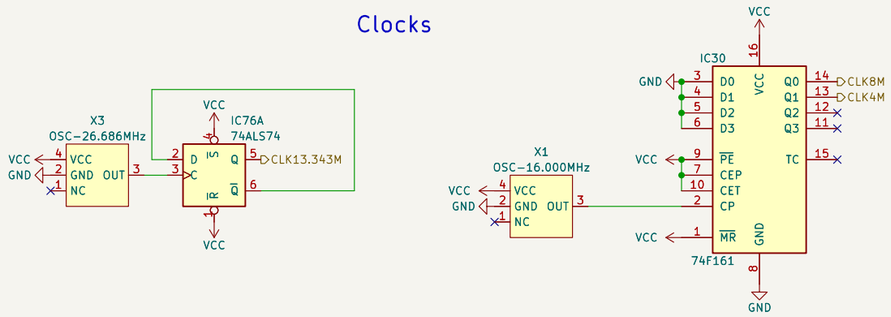
Title: Rainbow Islands Extra

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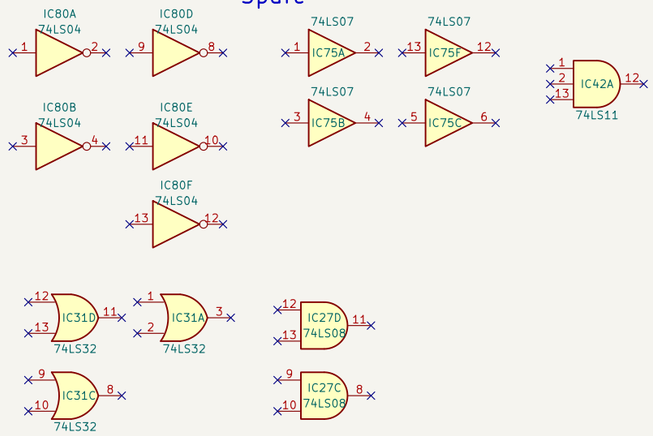
Watchdog and reset



Clocks



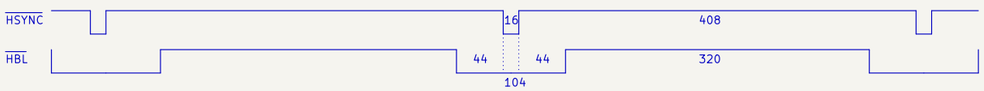
Spare



Horizontal and vertical synch timing diagrams

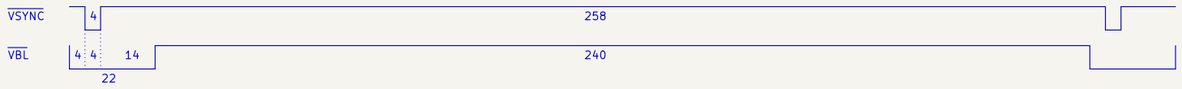
The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: f = 26.686MHz / 4 = 6.6715MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.



HSYNC and HBL
Frequency f = 6.6715MHz / 424 = 15.734670kHz.
Period T = 1/f = 63.5539us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.734670kHz / 262 = 60.05599Hz
Period T = 1/f = 1 / 15.734670kHz = 16.6511ms

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Sheet: /Misc/
File: misc.kicad_sch

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